

SANJULA BANDARA

Data Science Undergraduate | AI & GIS Intelligence Systems | Machine Learning | Geospatial Analytics

Homagama, Sri Lanka | 071 4280408 | sanjulabandara1113@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data Science undergraduate with GPA 3.7+ and a strong AI/GIS portfolio focused on Sri Lanka-specific operational intelligence systems. Built and deployed 5 end-to-end projects across maritime surveillance, terrain intelligence, disaster monitoring, wildlife conflict analytics, and transport safety using Python, Streamlit, Plotly, Folium, PyDeck, Pandas, and Scikit-learn. Seeking an internship in Data Science, GIS, AI monitoring systems, or IT & GIS research environments, with particular interest in defence-oriented geospatial intelligence, satellite/GIS systems, and machine learning applications.

TARGET ROLE KEYWORDS

Data Science | GIS Analytics | Machine Learning | Geospatial Intelligence | Operational Dashboards | Environmental Monitoring | Surveillance Analytics | AI-Based Decision Support | Python | Streamlit | Scikit-learn | Plotly | Folium | PyDeck | Data Cleaning | EDA | Model Training | Deployment

TECHNICAL SKILLS

Programming & Data	Python, Pandas, Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Data Analytics
Machine Learning	Scikit-learn, Random Forest, Classification Models, Feature Engineering, Label Encoding, Model Evaluation
GIS & Visualization	GIS Analytics, Folium, PyDeck, Plotly, Geospatial Dashboards, Heatmaps, Risk Maps, Operational Visuals
Deployment & Tools	Streamlit, Streamlit Cloud, Git, GitHub, GitHub Pages, Portfolio Website, Dashboard UI/UX

EDUCATION

NSBM Green University | Expected Graduation: November 2027

BSc (Hons) in Data Science

- Current GPA: 3.7+

G.C.E. Advanced Level - Engineering Technology Stream | Island Rank: 81 | Z-Score: 2.6673

- Results: ICT - A, Science for Technology - A, Engineering Technology - B

PORTFOLIO PROJECTS

Sri Lanka Maritime Surveillance & Coastal Intelligence Platform

Python, Streamlit, PyDeck, Plotly, Scikit-learn, GIS Analytics

Developed a flagship AI-powered maritime surveillance platform to monitor vessel activity, detect suspicious movement, analyze coastal threat zones, and predict maritime threat levels using machine learning.

- Designed a surveillance-operations interface with vessel intelligence feed, radar-style GIS visualization, coastal zone threat ranking, maritime simulation engine, and operational watchlist.
- Implemented ML-based threat classification using vessel type, speed, heading, suspicious activity score, signal loss, restricted-zone proximity, sea condition, and weather severity.
- Deployed live system and GitHub repository to demonstrate operational intelligence, anomaly detection, and coastal surveillance workflows.

Sri Lanka Landslide Terrain Intelligence System

Python, Streamlit, Folium, PyDeck, Plotly, Scikit-learn

Built a command-center-style terrain intelligence and landslide early-warning platform for Sri Lanka using GIS heatmaps, terrain threat corridors, simulation-based analysis, and AI terrain-failure prediction.

- Created dynamic emergency intelligence feed, terrain risk matrix, soil moisture trend analysis, national threat assessment, and infrastructure vulnerability analysis.

- Developed AI prediction engine using elevation, slope, rainfall, soil moisture, vegetation loss, historical landslides, river distance, and shelter availability.
- Improved project differentiation through operational UI/UX, simulation workflows, and geospatial heatmap intelligence layers.

Sri Lanka Flood & Rainfall Risk GIS Platform

Python, Streamlit, Folium, Plotly, Scikit-learn

Developed a disaster intelligence dashboard for rainfall monitoring, flood-prone district analysis, environmental risk scoring, and AI-based flood-risk prediction.

- Integrated GIS flood-risk mapping, rainfall trend analysis, district-level flood risk scoring, emergency priority zones, and downloadable processed data.
- Used rainfall, river level, soil saturation, population density, flood history, and drainage condition as predictive features.

Sri Lanka Human-Elephant Conflict Risk GIS System

Python, Streamlit, GIS Analytics, Scikit-learn

Built a wildlife intelligence dashboard for monitoring human-elephant conflict hotspots and predicting conflict-risk levels across Sri Lanka.

- Analyzed elephant sightings, crop damage reports, forest cover, water-source distance, population density, previous conflicts, and seasonal factors.
- Created district risk intelligence, seasonal conflict trends, emergency priority zones, and AI conflict-risk prediction.

Sri Lanka Road Accident Hotspot GIS Intelligence System

Python, Streamlit, Folium, Plotly, Scikit-learn

Created an AI-powered GIS dashboard for Sri Lankan road accident hotspot monitoring, district-level risk analysis, and accident severity prediction.

- Implemented interactive GIS mapping, district risk scoring, severity distribution, accident trend analysis, emergency priority zones, and ML-based severity prediction.

EXPERIENCE

Freelance Data Analytics Projects for A/L Tuition Teachers | 2024 - Present

Data analytics, dashboard development, reporting support

- Supported educational data analytics and reporting workflows for Advanced Level tuition teachers, including dashboard-style outputs and data summaries.
- Applied data cleaning, preprocessing, visualization, and reporting techniques to convert raw educational data into usable insights.
- Communicated requirements with non-technical users and delivered practical analytics outputs for decision-making and presentation purposes.

RESEARCH INTERESTS

Machine Learning, Deep Learning, GIS Analytics, Satellite and Geospatial Systems, Defence Systems, Surveillance Intelligence, Environmental Monitoring, Operational AI Decision Support.

SOFT SKILLS & LANGUAGES

Soft Skills: Communication, Presentation, Teamwork, Teaching, Announcing, Analytical Thinking, Problem Solving. | Languages: Sinhala, English.

CERTIFICATIONS

Currently planning to complete certifications in Machine Learning, GIS Analytics, Data Science, and AI-based operational intelligence systems.